Reading Both Halves of *Imvo*: What Modernising 1880s isiXhosa Does to Cross-Lingual Analysis

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Abstract

***Imvo Zabantsundu* (1884–1894) printed an isiXhosa and a smaller English section in every issue, so comparing them asks what one editor gave two readerships. We examine two choices usually treated as neutral: how language identity is removed from multilingual embeddings, and whether 1880s isiXhosa spelling is modernised first. We estimate LaBSE's language subspace only from translated pairs and measure the content gap as a section classifier's balanced accuracy above a translated-pair floor; on synthetic data this recovers the planted gap, while mean-centring reports zero. A lexicon-gated normaliser restores unwritten aspiration and joins separated prefixes, rewriting the names of peoples, places and God (*Tixo*→*Thixo*). [H1: the content gap on raw text, with CI.] [H2: whether modernisation changes the gap.] We recommend that analyses of historical African-language text report a normalisation-sensitivity check.**

Keywords: historical text normalisation; multilingual sentence embeddings; isiXhosa

Introduction

Imvo Zabantsundu, founded by John Tengo Jabavu in 1884, was the first African-owned and run newspaper in South Africa. It was written mainly in isiXhosa, but also printed an English translation of the editorial and, at times, other English articles, letters and advertisements (Schoots, Ngwendu, et al., 2023). Each issue therefore addressed two readerships under one editor. Comparing them with NLP tools invites "cleaning" the isiXhosa half first by modernising its spelling: the archive writes *Tixo*, *apa* and *umteto* where modern isiXhosa writes *Thixo*, *apha* and *umthetho*.

Research question. *When Imvo's isiXhosa and English sections from the same issue are placed in one multilingual embedding space, how distinguishable is their content, and how does modernising the isiXhosa spelling change that answer?* **H1:** with language identity removed, a classifier predicts a chunk's section above a translated-pair floor, i.e. the readerships received different content. **H2:** modernisation lowers the floor but leaves the gap unchanged; a changed gap would mean a preprocessing choice alters the conclusion.

Background The isiXhosa Intellectual Traditions archive was OCR-corrected by first-language isiXhosa speakers and deliberately not spell-checked, because the orthography of the period differs from present-day isiXhosa (Schoots, Ngwendu, et al., 2023). Historical normalisation is evaluated by accuracy or downstream scores (Bollmann, 2019), not by its effect on conclusions, which makes it a forking path (Gelman & Loken, 2013). Multilingual encoders such as LaBSE (Feng, Yang, et al., 2022) still encode language identity. It can be removed by subtracting each language's mean (Libovický, Rosa, et al., 2020) or by projecting out a low-rank subspace (Yang, Yang, et al., 2021; Ravfogel, Elazar, et al., 2020). For a linear probe, mean-centring also removes average content differences, the very thing we measure, so we estimate the subspace from translated pairs only.

Methods

Data. We use language-split text for 511 of the 517 Imvo issues in our copy (the archive lists 518) with $\geq$1,000 words per language: 3.9M isiXhosa and 1.5M English words, close to the archive's counts (Schoots, Ngwendu, et al., 2023). We repair OCR artefacts but never spelling, cut pages into 60-word chunks, and drop boilerplate ($\geq$30% of 5-word sequences recurring in $\geq$5 issues; 29.9% of isiXhosa and 7.9% of English chunks) and chunks in the wrong language. This leaves 45,465 isiXhosa and 22,927 English chunks, split by issue 357/77/77 into train/validation/test, stratified by year.

Modernisation. From 3.3M modern isiXhosa tokens **[verify]** (Vukuzenzele (Lastrucci, Rajab, et al., 2023), Wikipedia, OPUS-100 (Zhang, Williams, et al., 2020)) we build a frequency list f. *Aspiration* restores unwritten aspiration ($t, p, k, r, c, q, x \rightarrow +h$): w becomes its most frequent variant w' only if $f(w) \leq \tau_o$ and $f(w') \geq \tau_v$. *Prefix* joins prefixes written apart (*e Palamente* → *ePalamente*) only if the joined form, in any aspirated spelling, has $f \geq \tau_v$ (before a lowercase word, also five times the bare word's). (τ_o, τ_v) maximise $F_{0.5}$ on hand-checked validation words, as over-correcting an author is the costlier error **[tuned values]**.

Pairs. Chunks are embedded with LaBSE and multilingual-E5 (Wang et al., 2024). Mutual nearest neighbours across an issue's two sections are candidate translations; we keep LaBSE candidates with similarity $\geq$0.55 **[lowest threshold with validation precision $\geq$0.9]**, and repeat at $\geq$0.65. E5, whose similarities lie between 0.83 and 0.87, keeps the same share of its candidates.

Removing language, measuring content. Let P be the translated pairs in training issues. Because a pair shares its content, its difference is language:

$$D = \left[x_i^{\text{xh}} - x_i^{\text{en}}\right]_{i \in P}, \qquad \tilde{x} = x - V_k^{\top} V_k x, \tag{1}$$

where V_k holds the top-k right singular vectors of D. A class-balanced logistic-regression probe predicts each chunk's section from $\tilde{x}$. Its balanced accuracy on held-out pairs is the *floor* (language signal that survives), and the content gap is $\Delta = \text{BA}_{\text{all}} - \text{BA}_{\text{pairs}}$. $k \in \{0, 1, 2, 4, \ldots, 64\}$ is the smallest value whose validation floor is $\leq$0.55, and $C \in \{10^{-2}, \ldots, 10^{3}\}$ is tuned on validation. A cosine kNN probe repeats the measurement, as a linear probe cannot see differences that mean-centring leaves. For H2, variants reuse the raw pairs and k; we report paired bootstrap CIs over test issues (B=1000) and call a variant equivalent to raw if the 90% CI of its change in Δ lies within $\pm$0.02 (two one-sided tests, α=0.05). Alignment uses pairs matched by shared names and numbers **[hand-verified]**, which no embedding selected. On synthetic data with a planted gap δ, we compare against an oracle that sees content only, with clean and with 40% false pairs.

Results

The method recovers known gaps. On synthetic data, without language removal the probe separates translated pairs perfectly (floor 1.00, gap 0): it detects language, not content. Mean-centring reports a gap of 0.000 at every δ, as expected for a linear probe. The pair-based method tracks the oracle (Figure 1; 0.097 vs 0.099 at $\delta{=}0.4$) and, with 40% false pairs, underestimates rather than inflates it (0.045 vs 0.059).

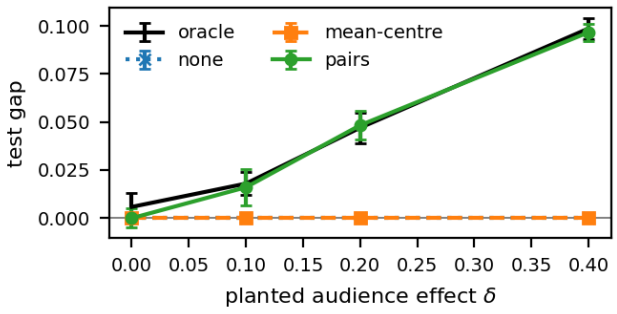


Figure 1: Synthetic validation (5 seeds; error bars = s.d.): only the pair-based method recovers the oracle gap; no removal and mean-centring both report 0.

What modernisation changes. With provisional thresholds, the full normaliser rewrites **[12.4%: update after re-run]** of the 3.7M alphabetic isiXhosa tokens, 7.8% by restoring aspiration (*apa*→*apha*, *pakati*→*phakathi*). **[Normaliser precision/recall on held-out words.]** Aspiration alone rewrites 149 likely proper names (capitalised in ≥90% of uses), including *Xosa*→*Xhosa*, *Tembu*→*Thembu*, *Mtata*→*Mthatha* and *Tixo*→*Thixo*. In Table 1, restoring aspiration cuts OOV from 31.5% to 24.5% while fertility barely moves: modernisation acts on vocabulary, not segmentation. isiXhosa needs about two-thirds more LaBSE subword tokens per word than English, in 1880s and modern text alike.

Table 1: OOV: share of tokens absent from the modern word list; LaBSE, XLM-R: subword tokens per raw word.

Text	OOV	LaBSE	XLM-R
Imvo isiXhosa, raw	31.5%	2.11	2.50
aspiration restored	24.5%	2.09	2.51
modern **[re-run]**	26.8%	2.12	2.52
Imvo English	–	1.26	1.38
Modern isiXhosa (Vukuzenzele)	–	2.18	2.92

Two readerships (H1) and the effect of modernisation (H2). Table 2 reports the probe on test issues ($[n]$ reference pairs, $k{=}$**[k]**). **[H1 in one sentence, with the 0.65-threshold and kNN gaps.] [H2 in one sentence: the verdict for each variant.] [Qualitative: the clusters most over-represented in the isiXhosa half, with one short quoted example.]**

Discussion

The control is the method. Mean-centring silently assumes that H1 is false; pair-based removal does not, and recovers planted gaps, conservatively when pairs are noisy. Since Imvo's English section centred on the translated editorial (Schoots, Ngwendu, et al., 2023), a positive gap should reflect what the English half left out, such as letters, local news and notices **[confirm with the clusters]**.

Table 2: Section probe on test issues (balanced accuracy); Δ = all − floor; change from raw with 90% CI; P@1 on embedding-free pairs.

Variant	All	Floor	Δ	$\Delta - \Delta_{\text{raw}}$	P@1
raw	**[]**	**[]**	**[]**	–	**[]**
modern	**[]**	**[]**	**[]**	**[CI]**	**[]**
aspiration only	**[]**	**[]**	**[]**	**[CI]**	**[]**
prefix only	**[]**	**[]**	**[]**	**[CI]**	**[]**
E5: modern	**[]**	**[]**	**[]**	**[CI]**	**[]**

Modernisation is a forking path. **[Interpret H2: safe for this question, or a conclusion that hinges on preprocessing.]** The ablation shows which rule drives this, and Table 1 why: modernisation acts on vocabulary, not on segmentation.

What "fixing" text costs. The normaliser rewrites the names of a people (*Xosa*), a community (*Tembu*), a town (*Mtata*) and God (*Tixo*) into today's forms. These are correct modern spellings, but they replace how the authors wrote their own names and beliefs, which the archive itself preserves (Schoots, Ngwendu, et al., 2023). We recommend that analyses of historical African-language text report a *normalisation-sensitivity* check: results on raw and on normalised text.

Limitations. Pairs are concentrated in some genres **[genre mix]** and 43% come from 1885–86. Labelling was done by **[the team's isiXhosa reading level]**. Equal-length chunks hold more content in isiXhosa than in English, which caps pair similarity. Residual OCR noise is unmeasured (Schoots, Ngwendu, et al., 2023). We measure content, not editorial intent.

Conclusion

Removing language with translated pairs keeps the content difference that mean-centring erases, and spelling modernisation should be reported and ablated, not treated as cleaning. **[What H1 and H2 showed.]**

Contribution Statements (Not in Page Limit)

We (Lily de Melo, Tawananyasha Kadango and Sally Mohlakola) worked together fairly on this project. **[Paste the final contribution statement; all group members must agree with it.]**

References

Bollmann, M. (2019). A large-scale comparison of historical text normalization systems. In *Proceedings of NAACL-HLT.*

Feng, F., Yang, Y., Cer, D., Arivazhagan, N., & Wang, W. (2022). Language-agnostic BERT sentence embedding. In *Proceedings of ACL.*

Gelman, A., & Loken, E. (2013). *The garden of forking paths* (Tech. Rep.). Columbia University.

Lastrucci, R., Rajab, J., Shingange, M., Njini, D., & Marivate, V. (2023). Preparing the Vuk'uzenzele and ZA-gov-multilingual South African multilingual corpora. In *Proceedings of the fourth workshop on resources for african indigenous languages (RAIL 2023)* (pp. 18–25).

Libovický, J., Rosa, R., & Fraser, A. (2020). On the language neutrality of pre-trained multilingual representations. In *Findings of EMNLP*.

Ravfogel, S., Elazar, Y., Gonen, H., Twiton, M., & Goldberg, Y. (2020). Null it out: Guarding protected attributes by iterative nullspace projection. In *Proceedings of ACL* (pp. 7237–7256).

Schoots, J., Ngwendu, A., De Wet, J., & Muftic, S. (2023). IsiXhosa intellectual traditions digital archive: Digitizing isiXhosa texts from 1870–1914. In *Proceedings of the fourth workshop on resources for african indigenous languages (RAIL 2023)* (pp. 54–64).

Wang, L., et al. (2024). *Multilingual E5 text embeddings: A technical report.* (arXiv:2402.05672)

Yang, Z., Yang, Y., Cer, D., & Darve, E. (2021). A simple and effective method to eliminate the self language bias in multilingual representations. In *Proceedings of EMNLP* (pp. 5825–5832).

Zhang, B., Williams, P., Titov, I., & Sennrich, R. (2020). Improving massively multilingual neural machine translation and zero-shot translation. In *Proceedings of ACL.*

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    Answer: [NA]

    Justification: No human subjects: the texts are newspapers from 1884–1894, and the archive's authors note that no ethics review was needed because the materials discuss no living subjects (Schoots et al., 2023).

    Guidelines:

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Wits Faculty of Science AI Ethics Statement

[Faculty of Science AI Ethics Form.]